

MATHEMATICS FOR AI

Assignment 2

Please take care about your hand written picture. Make sure that your solutions are detailed enough since in case of mistakes/errors we can grade them better. Do not forget to write your name on the top right for all of the paper sheets.

Problem. Let us consider the following neural network architecture:

- ◇ Input layer
- ◇ Hidden layer 1 $\sim$ ReLU activation
- ◇ Hidden layer 2 $\sim$ Sigmoid activation
- ◇ Output layer $\sim$ softmax activation

Specifically let us consider the choices

$$\diamond x = (1 \ 0 \ -1 \ 2 \ 1 \ 0 \ -2 \ 1)^T$$

$$\diamond A^{(1)} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}, \quad b^{(1)} = (0 \ 1 \ -1 \ 0)^T$$

$$\diamond A^{(2)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \quad b^{(2)} = (0 \ 0 \ -1 \ 0 \ 0 \ 0)^T$$

$$\diamond A^{(3)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \quad b^{(3)} = (0 \ 0 \ 0 \ 0 \ 0)^T$$

Solve the following tasks:

1. Determine the final output vector. Where necessary please represent the solutions to two decimal places.
2. Represent your neural network as a composite function.